Final Report: S.C.R.A.P.

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1 Mission Concept & Objectives

1.1 Mission Overview

Low Earth Orbit is facing a growing issue pertaining to dysfunctional satellites and rogue objects. These threaten the safety of other LEO satellites since a collision could cascade into an increasingly hard to control quantity of space debris. SCRAP aims to systematically remove objects from orbit. By choosing high-value targets, missions would have a high return-on-investment in terms of reducing risk to other satellites.

Removing a target from orbit consists of two major processes: de-tumbling, and de-orbiting. De-tumbling is a necessary task that removes the barrier to engagement that is uncontrolled rotation. De-orbiting will be done with a passive drag device that significantly increases the decay rate that the target experiences. This will decrease the de-orbiting time from an order of magnitude of years to months. These processes will be repeated for multiple targets in one mission, increasing financial viability and impact of each mission.

1.2 Economic Model

1.2.1 Market Summary

The commercial Low Earth Orbit (LEO) environment is rapidly reaching a critical saturation point. Driven by mega-constellations like Starlink and Project Kuiper, the LEO satellite market is projected to surpass $24 billion by 2032. However, this exponential growth brings about the systemic physical and financial threat of Kessler Syndrome. Operators are now having to deal with increased insurance premiums and tightened regulatory constraints. SCRAP is engineered to directly address this bottleneck in orbital life-cycle management by providing a scalable, autonomous active debris removal service.

1.2.2 Regulatory Impetus

Enforced as of late 2024, the FCC strictly mandates that domestic and market-access LEO satellites must undergo post-mission disposal within five years of operational end-of-life. Despite recent lobbying pushback from major constellation operators to relax these constraints, the rule remains a hard baseline, heavily penalizing non-compliance with the denial of future launch licenses.

ESA has instituted a comprehensive charter aiming to eliminate all debris creation in "valuable" orbits by 2030. To comply, new missions must demonstrate a guaranteed disposal success rate exceeding 90%, creating a captive market for third-party disposal services.

1.2.3 SCRAP Architecture

Unlike legacy capture concepts that use harpoons, nets, or complex robotic grappling, the SCRAP architecture will use a contactless high-temperature superconductor (HTS) electromagnet system for magnetic eddy current de-tumbling. Projecting a magnetic field towards the spinning debris will induce eddy currents in the conductive aluminum and titanium structures of target satellites that will generate a braking torque.

This allows SCRAP to bleed off angular momentum from uncooperative, rapidly tumbling targets from a safe standoff distance before executing a rendezvous. This eliminates the risk of blunt force fragmentation during capture, a critical point of failure for current proposed active debris removal architectures.

1.2.4 Competitive Landscape

The space traffic management regulatory apparatuses are validating our business model, though leading competitors remain constrained by complex kinetic capture mechanisms.

- **Astroscale:** The current market leader. They are executing missions like ADRAS-J (JAXA upper stage inspection) and the ELSA-M servicer targeting OneWeb satellites. They are also moving into refueling with the APS-R mission for the US Space Force. However, their primary capture mechanisms rely heavily on cooperative targets with pre-installed magnetic docking plates or complex, high-risk robotic arms.
- **Starfish Space:** Maturing their Otter servicing vehicle, recently partnering with NASA for an orbital debris inspection study. While their autonomous rendezvous and proximity operations (RPO) are advanced, they still require physical grappling to secure uncooperative mass.

1.2.5 Dual Use Commercialization Strategy

SCRAP's contactless de-tumbling mechanism inherently supports a highly lucrative, dual-use go-to-market strategy:

- Civil / Intergovernmental (ESA / UN COPUOS): Positioning SCRAP as a neutral, "global good" utility. Securing public cooperation contracts under the ESA Zero Debris framework provides stable, non-dilutive foundational revenue and establishes vital flight heritage.
- US Department of Defense (Space Force): The ability to execute contactless attitude control on uncooperative objects has profound implications for Space Domain Awareness (SDA) and exerting hard power in space. SCRAP can be pitched to the DoD as a platform capable of safely neutralizing hostile or compromised assets without generating additional orbital debris, aligning directly with USSF directives for "dynamic space operations."

1.3 Scientific and Engineering Objectives

SCRAP's success depends on the team's ability to accomplish the following objectives. These objectives, when completed, will establish a unique service that can satisfy our economic model.

- Identify high-value candidates for de-orbiting.
- Use a computer vision software to determine the angular velocity vector of a target using video feed.
- Develop a system capable of producing a large B-field that can be positioned in an optimal location for de-tumbling the target.
- Find an adhesive or attachment mechanism for attaching the GOLD system to the target.
- Determine the radius needed for the GOLD system to produce the necessary drag for de-orbiting.
- Generate and store enough energy to power subsystems.

2 Requirements Definition & Compliance

2.1 Mission-Level and Subsystem Requirements

1. The system shall de-orbit defunct or decommissioned satellites from Low-Earth Orbit safely and reliably without producing additional debris.

   The overall goal of reducing space clutter in LEO, which includes both debris and defunct satellites, has undergone refinement in the scope of an economically feasible mission architecture. Essentially, it is impossible to target all space debris, even within certain altitude bands and mass classifications. Thus, it is important to define the candidates for removal from LEO. Fiscally, it makes the most sense to target decommissioned or dysfunctional satellites, since the growing number of satellites in LEO is likely to become heavily regulated in the near future. The service that SCRAP could provide would then likely be viable in the growing space market for private space to avoid fines or for government agencies to clean up LEO in general. The industry standard for self-removal of satellites is de-orbit operations, and it makes sense for our system to do that same.

2. The de-tumbling system shall de-tumble satellites rotating in excess of 5 rad/min to a maximum angular velocity of 5 deg/sec within 2-weeks to aid in de-orbit operations.

   De-tumbling was derived from the parent requirement since de-orbit facilitation will require interfacing with the candidate object to attach the drag system. Accomplishing this requires matching the candidate object's movement with the chaser satellite, which is difficult to achieve in objects with high tumble rates. Seeing as tumbling objects are prevalent and prevent safe and reliable proximity operations, it is imperative that a dedicated de-tumbling system is available to widely increase the total candidate gamut for de-orbit.

3. The system shall operate autonomously throughout constellation de-orbit operations while also allowing for reliable data uplink and downlink during idle operations.

It is imperative that the satellite be capable of performing de-orbiting operations autonomously. Proximity operations are dangerous and require a robust system, and continuous remote control is infeasible for the projected mission duration.

4. The propulsion system shall maintain station-keeping in-between and attitude control during constellation de-orbit operations.

   A propulsion system must provide the system with enough delta-V for target de-orbit volume on a case-by-case basis, but with the capability of remaining in an orbit until candidates are identified. This likely entails the use of both electric and monopropellant propulsion systems.

5. The satellite bus shall provide necessary power and power management for components and subsystems to facilitate continued de-orbit operations throughout the system mission lifetime.

   The de-tumble system limits the satellite bus budget and as such, careful consideration must be made to the power budget, allocating enough to effectively facilitate de-tumbling operations without sacrificing reliability of the rest of the satellite to maintain the overall system operability.

These high-level parent requirements for essential subsystems are just that: high-level. The focus of necessary trade studies and subsystem optimization will drive additional, derived children requirements. As the team moves forward and subsystems require more extensive scrutiny, more comprehensive requirements shall be implemented to help drive development of the system.

2.2 Science Traceability Matrix

Mission Goal	Mission Objectives	Operational Requirements		Subsystem Requirements	Mission Reqiurements
		Physical Parameters	Observable		
Develop a satelite capable of reducing debris in LEO by rendezvousing with targets, de-tumbling them, and de-orbiting them.	Determine location and orientation for accurate menuevers.	ECI Position	GPS Transmission	LEO capable GPS receiver with an update rate > 1 Hz.	Onboard storing of GPS and comparison to propogated position.
		Acceleration	Voltage produced by IMU	IMU with a 3-axis accelerometer and a 3-axis gyroscope capable of > 500 sps.	Kalman filtered propogation for position between GPS transmissions.
		Attitude	Visual star orientation	Commercial space-grade star tracker	Must face towards open space.
	Determine the target state	Distance to target	Time taken by beam of light to reflect from surface	Time-of-flight sensor with an accuracy of < 0.5% of range.	Used in comparison with CV generated distance.
		Target rotational state	Change in pixel location in time series of visual 2D masks	Visible spectrum camera capable of 120 fps at 1080p. Onboard FPGA for computer vision modeling.	Inform position of magnet for optimal de-tubling.
		Target surface temperature	Infrared radiation emitted by surface	Low resolution infrared spectrum camera.	Known relationship between IR reading and temperature.
	Induce eddie currents in target to reduce angular velocity	Magnetic field induced	Current passed through coil	20-turn YBCO superconducting pancake coil with a 10 cm diameter capable of producing a 0.15 mT field at a 2 m standoff.	Placed in optial position relative to target for detumbling.
	Use a three stage adhesive to secure GOLD to target	Shear force applied to Van der Waals based dry adhesive	Release of radial spring	Mechanism capable of storing and releasing a radial spring with 1.9 cm of displacement.	Productive surface contact > 75% of adhesive pad.
		Heat up electrothermal dry adhesive	Discharge current from capacitor	RC circuit based heating element capable of reaching 20 C for 60 seconds.	Capacitor discharged at the correct time after Van der Waals
		Application of thermoplastic adhesive	Force applied by release of pressurized gas	Pressure system made up of a two chambers and a valve capable of containing CO2 and producing 14 kPa.	Last adhesion, strong enough to hold onto drag force generated by GOLD.
	Control pressure inside balloon	Gas particles released into balloon	Time since beginning of sublimation	Record of time.	Understanding of sublimation rate of material.
		The balloon being inflated	Current passed through limit switch	Embedded limit switch on the "end" of the sublimation material.	Placement to avoid false negatives.

Figure 1: Science Traceability Matrix

3 Concept of Operations

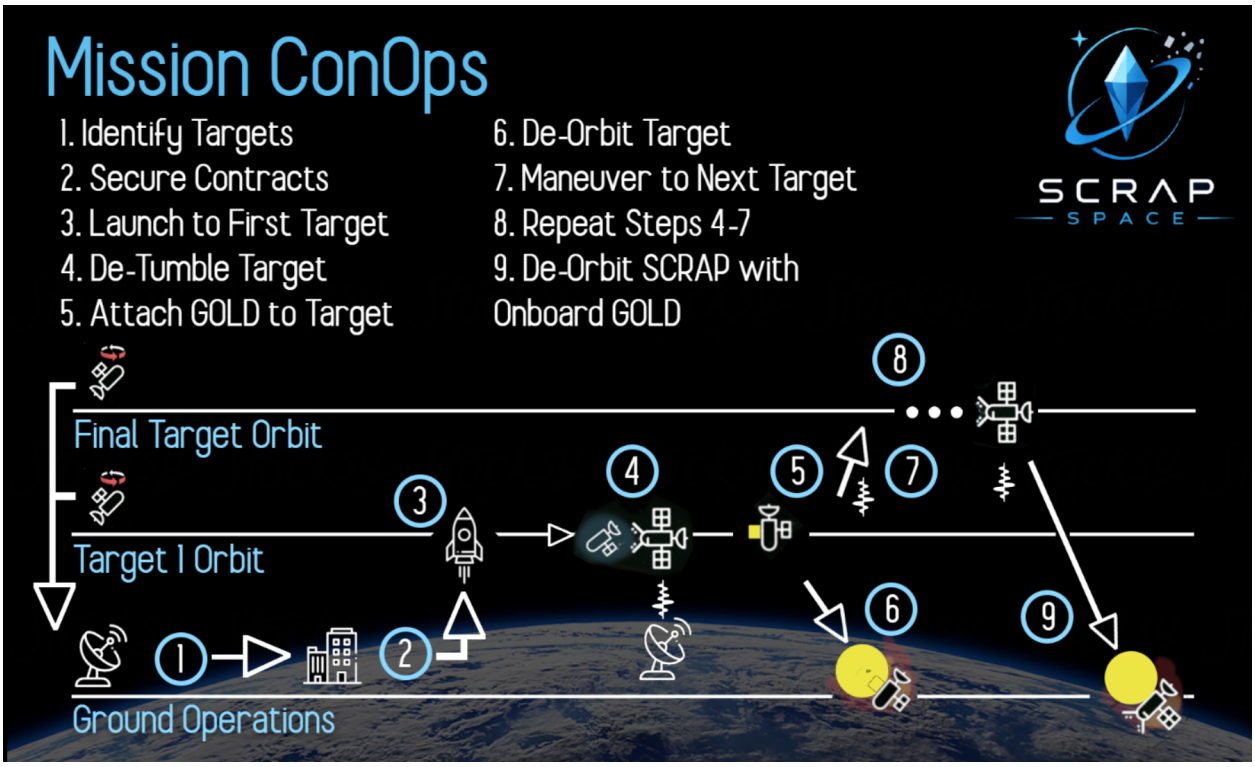


Figure 2: Concept of Operations diagram labeling the modes of operation for a SCRAP mission.

1. Candidate Selection
   First, high value candidates must be selected. This is critical for mission viability and financial support.

2. Commercial Realization
   After targets are identified, the mission must be realized commercially. This involves selling the service to parties of interest who will either face legal repercussion for a failed satellite or collision risk with uncontrolled space objects.

3. Launch
   After the targets are selected and the mission is funded, the spacecraft will launch.

4. De-Tumble Target
   Upon rendezvous the first goal is de-tumbling the target. After characterizing the rotation of the target, the electromagnetic de-tumbler is positioned to the optimal location to apply a torque against the angular momentum vector.

5. De-Orbit Target
   After de-tumbling, the GOLD system is attached, and the target is de-orbited. The GOLD system increases the drag of the target and reduces the de-orbit time to an order of weeks.

6. Next Target
   Once the GOLD system has been attached a target, the SCRAP spacecraft's job is complete.

The next step is maneuvering to the next target and repeating steps 4-6 until the mission is over.

7. De-Orbit
   After the mission is complete, the SCRAP satellite must de-orbit itself in order to not introduce more debris into LEO.

3.1 Operational Constraints

Altitude:

Due to a reliance on passive drag to de-orbit targets, there is an upper limit of the altitude of operation. Models currently show effective de-orbiting at altitudes as high as 700 km. However, the size of the drag device rapidly increases as altitudes increase further and thus, a ceiling of 500 km is chosen to minimize drag device mass. This will be further explained in the GOLD system section of the report.

Satellite Selection:

Satellite selection has an unfathomable number of potential considerations and therefore are abstracted away significantly. Ultimately specific selection will be heavily determined by negotiations with individual private companies OR with NASA/ESA or other governing bodies.

Magnet:

Since our de-tumbler will be using a high-temperature superconductor (HTS), quenching (a rapid loss of superconductivity and dissipation of heat) is the single largest concern when operating a superconductor. There are maximum current, temperature, and B field thresholds that, if surpassed, would cause catastrophic quenching that would melt the electro-magnet and very possibly also a large part of the spacecraft. Operating superconductors on Earth is difficult because of the cooling requirements, but in space we can take advantage of the low ambient temperature ($\sim 4°$K). In general, as temperature increases, the maximum B field and current density decrease that a Type II superconductor can withstand before quenching. Thus, for a given current density, as long as we maintain the system at some safety margin below the critical temperature, we will be able to sustain the magnet with no power input.

Attachment:

The drag device will need to engage with the target with an adhesive pressure on the order of 50 kPa. The nature of dry adhesives requires that the attachment surface is relatively smooth. It is also important that physical interface to attach the drag package is done in the shortest possible duration to minimize inertial momentum transfer from the chaser satellite to the candidate object.

Collision Risk:

The GOLD system drastically increases the cross sectional area of the target. This is the driver of the increased drag, so it cannot be avoided. If an object's necessary increase in cross sectional area risks collision with other satellites during de-orbit, the object cannot be targeted.

3.2 Rendezvous and Proximity Operations (RPO)

4 Subsystem Design

4.1 De-tumble System

4.1.1 Selection

Candidate selection is handled by a custom Rust-based pipeline that ingests and cross-references multiple satellite catalogs: Space-Track's SATCAT and GP databases, ESA's DISCOSweb for physical properties like mass and cross-sectional area, and the GCAT historical catalog. These sources are merged on NORAD catalog ID to build a unified database of LEO objects.

There are five components to the model:

- **Mass/Size (25%):** Log-normalized mass with a cross-sectional area bonus from DISCOSweb data. Larger objects produce more secondary debris upon fragmentation.
- **Collision Risk (35%):** Computed from the number of close approaches within a 72-hour propagation window, the maximum relative velocity of those approaches, and the minimum miss distance. A spatial cell hashing algorithm identifies conjunction events across the full catalog.
- **Decay Timeline (25%):** Objects that will not decay naturally within 10 years receive the highest scores. Objects already on a short decay trajectory are deprioritized since they will re-enter on their own.
- **Object Type (10%):** Rocket bodies score highest (1.0), followed by debris (0.5) and payloads (0.3).
- **Country/Operator (5%):** Configurable geopolitical weighting for mission planning flexibility.

An altitude band multiplier further scales the base score by orbital congestion. The 450–600 km megaconstellation band receives a 2.0x multiplier, the 600–800 km sun-synchronous band gets 1.8x, and other altitude bands are weighted accordingly.

Decay predictions use several numerical propagators with configurable atmosphere models and J2 or full 20×20 gravity fields. The system outputs a ranked list of candidates. In a full catalog run of roughly 29,000 tracked LEO objects, we identified over 1,000 close-approach events and ranked candidates by total weighted score out of 100.

4.1.2 Route Selection

Route selection builds on the ranked candidate list by determining which objects can be visited within a single mission and in what order. Rather than simply choosing the highest-scoring objects independently, the planner evaluates the orbital cost of transferring from the parking orbit to a candidate target and from each serviced target to the next. Candidate objects are first filtered to those with valid TLE data, within the allowed altitude band, and not already decayed. Their orbital geometry is then evaluated at the planned launch epoch so that nodal drift is handled consistently across the full candidate set, preventing the route design from being based on stale catalog geometry. For rapid screening, each candidate leg is assigned a first-order maneuver cost made up of an altitude-change term, a plane-change term, and a fixed phasing allowance,

$$\Delta v_{\text{leg}} \approx \Delta v_H + \Delta v_{\text{plane}} + \Delta v_\phi. \tag{1}$$

The altitude-change term is estimated with a Hohmann-style transfer,

$$\Delta v_H = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1+r_2}}-1\right)+\sqrt{\frac{\mu}{r_2}}\left(1-\sqrt{\frac{2r_1}{r_1+r_2}}\right), \tag{2}$$

while the relative plane mismatch is approximated from inclination and RAAN as

$$\Delta\theta = \cos^{-1}(\cos i_1 \cos i_2 + \sin i_1 \sin i_2 \cos(\Omega_2 - \Omega_1)), \qquad \Delta v_{\text{plane}} \approx 2\sqrt{\frac{\mu}{\max(r_1, r_2)}}\sin\left(\frac{\Delta\theta}{2}\right). \tag{3}$$

A corresponding time of flight is estimated from the half-period of the transfer ellipse with a small phasing bias,

$$t_{\text{leg}} \approx \pi\sqrt{\frac{\left(\frac{r_1+r_2}{2}\right)^3}{\mu}} + 0.25\,T_2. \tag{4}$$

These low-cost estimates allow the software to reject infeasible legs early based on total mission duration, total delta-v budget, and optional plane-change limits. After screening, the route is assembled using either a greedy search or a beam search, both of which favor targets that return the greatest value per unit maneuver cost. The local ranking metric is

$$J_j = \frac{S_{\text{eff},j}(t)}{\Delta v_j}, \tag{5}$$

where $S_{\text{eff},j}(t)$ is the value of target j at the expected arrival time. In the default time-decay model,

$$S_{\text{eff},j}(t) = S_j \max\left(0, \frac{D_j - t}{D_j}\right), \tag{6}$$

with S_j the original value score and D_j the predicted days to natural decay. This prevents the route planner from over-prioritizing objects that are likely to re-enter on their own before rendezvous occurs. Once a leg survives approximate screening, the planner can either keep the first-order estimate or refine the transfer using impulsive, low-thrust, or Lambert-based models. The final output is a ranked set of feasible multi-target routes, each reported with total delta-v, mission duration, and adjusted score for mission-level trade studies.

4.1.3 Rotation Determination

Determining the angular velocity vector of a tumbling target is necessary for positioning the electromagnet optimally during de-tumbling. This is done using a vision pipeline built on classical optical flow techniques rather than deep learning, which keeps computational requirements low enough for onboard processing.

The pipeline has three stages. First, each video frame is segmented using Canny edge detection followed by dilation to produce a binary mask which separates the target from the space background. Second, sparse features are detected on the target surface using the Shi-Tomasi corner detector and tracked between consecutive frames with Lucas-Kanade optical flow.

Third, the tracked 2D feature displacements are decomposed into translational and rotational components. The translational component from the chaser spacecraft's own motion is subtracted using known velocity (onboard IMU). The residual motion is then fit to a rigid-body rotation model via least squares:

$$\dot{u} = f_x(-xy\omega_x + x^2\omega_y - y\omega_z), \quad \dot{v} = f_y(-y^2\omega_x + xy\omega_y + x\omega_z) \tag{7}$$

where f_x, f_y are the camera focal lengths and $\omega_x, \omega_y, \omega_z$ are the angular velocity components being estimated. With n tracked features this produces a $2n \times 3$ overdetermined system solved each frame. A 5-frame moving average smooths the output.

To validate this approach, Blender was used to render synthetic satellite tumble sequences at known angular velocities. On a 600-frame test at 15 FPS with per-axis angular rates ranging from 1.5 to 11.25 degree/s, the system tracked features through all frames and produced angular velocity estimates that followed the ground truth. Current work focuses on testing with more realistic satellite geometries rendered from more realistic lighting and characterizing accuracy at the lower angular velocities expected after partial de-tumbling.

4.1.4 Magnetic De-tumbler

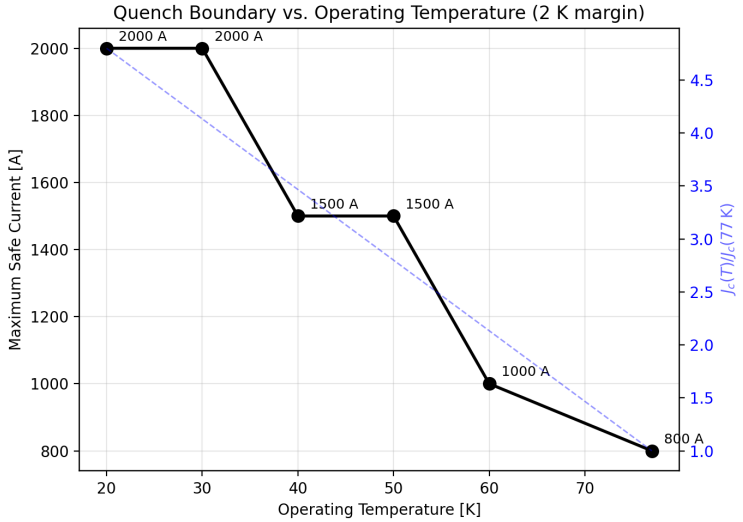


Figure 3: Maximum safe operating current as a function of temperature for the YBCO pancake coil. At 77 K the coil supports 800 A, cooling to 20 K extends this to 2,000 A. The dashed line shows the relative increase in critical current density $J_c(T)/J_c(77\,\mathrm{K})$.

The magnetic de-tumbling system uses a YBCO (Yttrium Barium Copper Oxide) high-temperature superconductor pancake coil to generate a strong magnetic dipole. When this field is projected toward the tumbling satellite, eddy currents are created in the conductive aluminum and other structures of the target. This produces a braking torque that opposes the rotation. Our satellite never has to come into contact with the target satellite.

The coil itself is a 20-turn pancake made from a YBCO tape which is a superconductor on a Hastelloy substrate with a copper stabilizer. YBCO has a critical temperature of 92 K, meaning it superconducts above the 4 K space background. The engineering constraint is operating temperature versus current capacity. A coupled electromagnetic-thermal finite element model was built to map out this space. The EM solver uses an H-formulation with E-J power law resistivity, and the thermal solver accounts for Joule heating, radiative cooling to space, and cryocooler contact cooling. Parameter sweeps using a faster analytical model confirmed that at 77 K, the coil safely handles up to 800 A, while cooling to 20 K with a cryocooler extends this to over 2,000 A. The critical current density follows a Kim-like model, $J_c(T, B) = J_{c0}(1 - T/T_c) \cdot B_0/(B_0 + |B|)^{\alpha}$, where both temperature and self-field suppress J_c. Lower operating temperatures increase J_c and provide

substantially more margin before quench. Current ramp rate also matters because slower ramps spread out heat dissipation and delay quench onset by up to 75x compared to fast ramps.

For positioning the coil, a reinforcement learning agent was modified from existing research using the Deep Deterministic Policy Gradient (DDPG) algorithm.

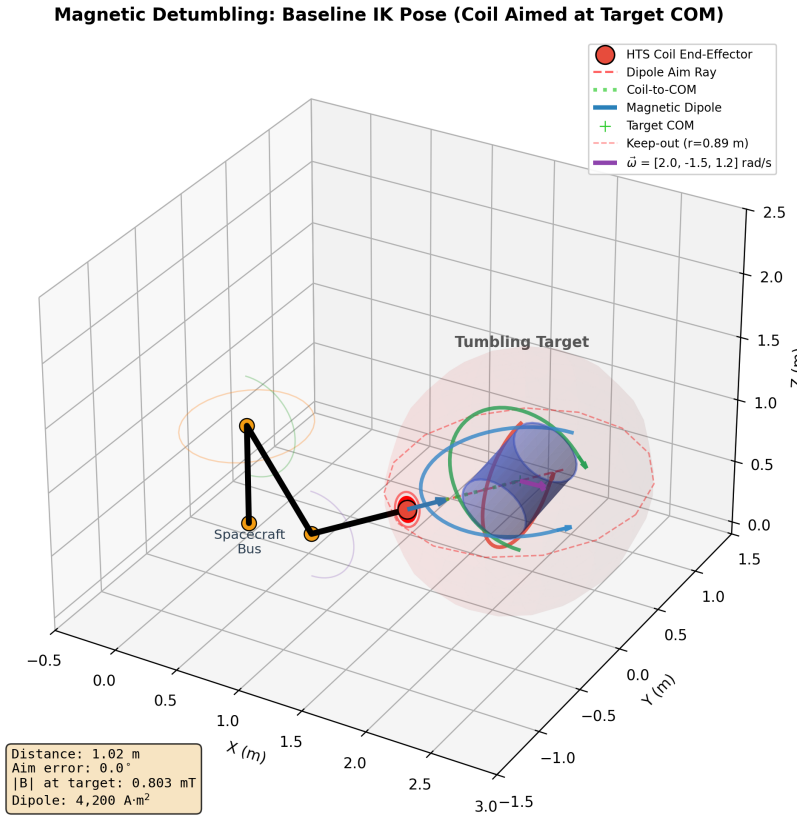


Figure 4: Baseline inverse-kinematics pose of the 3-DOF robotic arm with the HTS coil end-effector aimed at the tumbling target's center of mass.

The simulation environment models a 3-DOF robotic arm with the superconducting coil at its end-effector, producing a magnetic dipole moment of 4,200 A·m^2. The agent observes the target's angular velocity, the end-effector position, and the magnetic field at the target, and outputs incremental position commands for the arm tip. The induced torque on the target is modeled as $\boldsymbol{\tau} = (\boldsymbol{\chi} \otimes (\boldsymbol{\omega} \times \mathbf{B})) \times \mathbf{B}$, where $\boldsymbol{\chi}$ is a magnetic tensor representing the target's conductive structure.

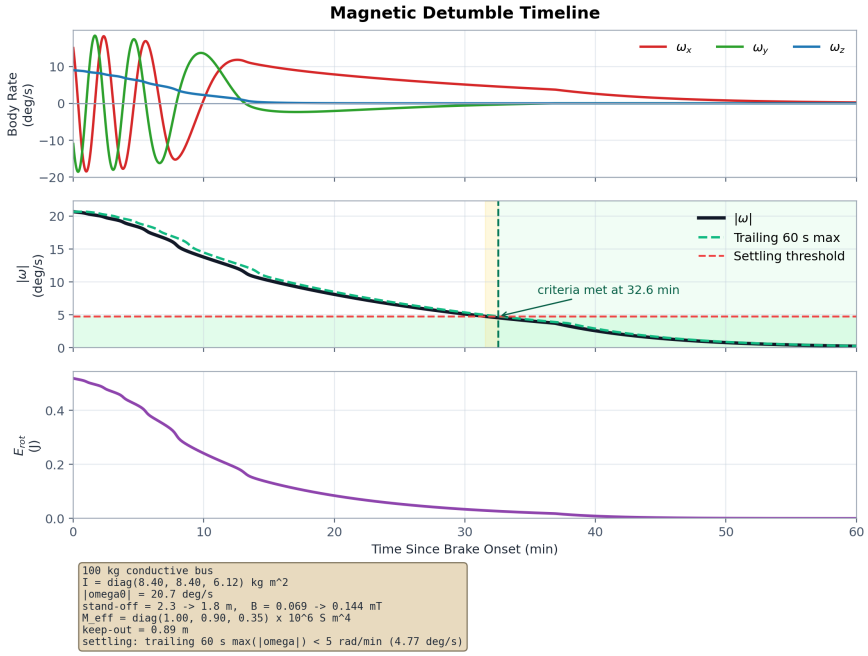


Figure 5: Simulated magnetic detumble of a 100 kg conductive target. Top: per-axis angular velocity components. Middle: angular speed magnitude with 60 s trailing maximum and the 5 rad/min settling threshold. Bottom: rotational kinetic energy. The settling criterion is met at 32.6 minutes.

4.1.5 De-Tumbling Trade Study

As an alternative to laser ablation and kinetic capture methods, we evaluated contactless magnetic eddy current detumbling. This method utilizes a magnetic (B) field generated by the servicer satellite to induce electromotive forces (EMFs) in the conductive structures of the tumbling target. According to Faraday's law of induction and Ohm's law, the time-varying magnetic flux experienced by the rotating target induces eddy currents within its conductive chassis. These currents generate their own secondary magnetic fields, which interact with the primary field to produce a braking torque that directly opposes the target's rotation.

Instead of relying on the computationally heavy Finite Element Method (FEM) to solve the Laplace equation with Neumann boundary conditions at each time step, the torque can be calculated using a "Magnetic Tensor" (M). This tensor depends solely on the physical properties (such as geometry and conductivity) of the target body and only needs to be computed once. The Eddy Current Torque ($\mathbf{T}_{eddy}$) is then calculated using the following formula:

$$T_{eddy} = (M\Omega) \times B \tag{8}$$

where $\Omega = \omega \times B$, with all quantities expressed in the target-fixed frame.

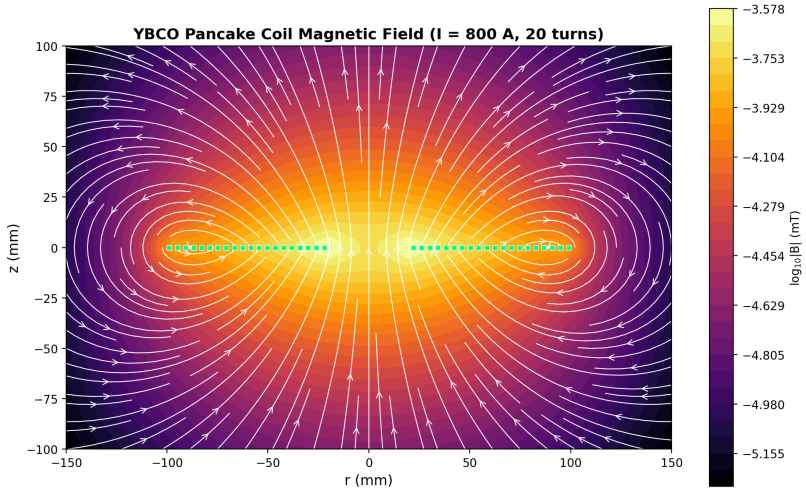


Figure 6: Cross-sectional magnetic field map of the YBCO 20-turn pancake coil at 800 A. Color shows $\log_{10}|\mathbf{B}|$ in mT with streamlines indicating field direction. Cyan dots mark individual turns.

To generate the needed magnetic field strength without incurring the massive power, mass, and heating penalties of traditional copper electromagnets, the SCRAP architecture employs high-temperature superconductor (HTS) electromagnets, specifically Yttrium Barium Copper Oxide (YBCO) pancake coils. When cooled below their critical temperature (a function of current density and B field strength), YBCO coils exhibit near-zero DC resistance, allowing for extremely high current densities with low long-term power draw. The artificial magnetic field $\mathbf{B}$ generated by this HTS magnet can be modeled as a dipole:

$$\mathbf{B} = \frac{\mu_0 n I A}{4\pi} \left(\frac{3\mathbf{r}(\boldsymbol{\mu} \cdot \mathbf{r})}{r^5} - \frac{\boldsymbol{\mu}}{r^3} \right) \tag{9}$$

where μ_0 is the vacuum magnetic permeability, n is the number of windings, I is the coil current, A is the inner coil area, $\mathbf{r}$ is the relative distance vector between the magnet's center and the target's center of mass, and $\boldsymbol{\mu}$ is the unit magnetic moment.

Furthermore, the magnitude of the eddy current torque is heavily dependent on the angle α between the magnetic field vector and the angular velocity vector (for certain geometries, with certain rotational velocity vectors such as those used in our trade study simulations). The torque magnitude is proportional to $\sin(\alpha)$, meaning the braking force is mathematically maximized when the applied magnetic field is kept strictly perpendicular to the RAV.

Our trade study comparing HTS eddy detumbling to the laser ablation yields stark contrasts, primarily in power consumption and system predictability:

4.1.6 Proximity Operations

The operational envelope for the HTS eddy detumbler is bounded primarily by the $1/r^3$ spatial decay of the magnetic dipole field, and the braking torque scales as $T_{eddy} \propto 1/r^6$. Thus the system is most effective at small separation distances, though we must operate it outside of some safety envelope so we do not collide with the rotating target. The electromagnet must be able to generate sufficient magnetic field strength ($> 100\,\mu$T) at the target's center of mass. This requires robust autonomous rendezvous and proximity operations (RPO) capabilities to maintain station-keeping during the detumble phase, also due to the transfer of angular momentum to the servicer satellite.

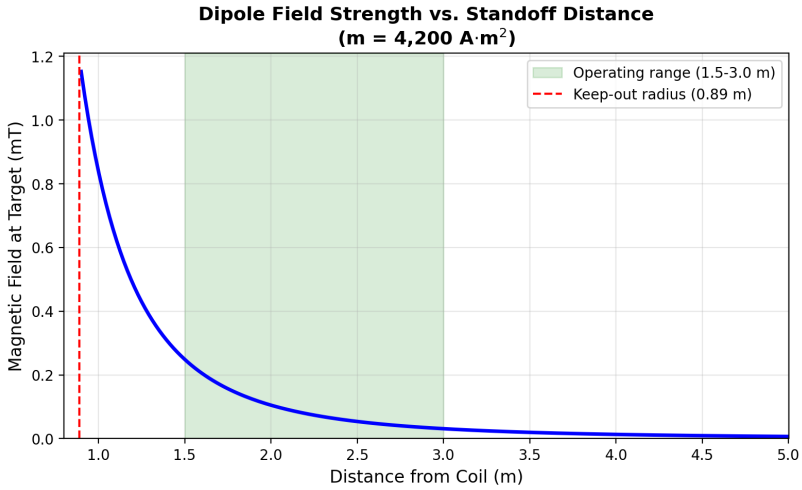


Figure 7: On-axis magnetic field strength at the target as a function of standoff distance for a 4,200 A·m^2 dipole. The green band marks the 1.5–3.0 m operating range; the red dashed line is the 0.89 m keep-out radius.

4.1.7 Power Draw and De-tumble Rates

This is the primary driver for selection. To detumble a 100 kg object in one day, laser ablation requires an unsustainable power draw on the order of 140 kW. In contrast, once the HTS electromagnet is energized, it operates with essentially zero power loss. The system's active power budget is entirely dominated by the any active cooling systems required to maintain the YBCO coils at operational temperatures ($<\sim$ 70 K). Modern space-rated cryocoolers draw on the order of 150 to 300 W, and we believe we can eliminate the need for one entirely via inductive ramp-up and thermal shielding. The detumbling process is considered successful when some success criterion (debris rotational energy or speed) ($E_r = \frac{1}{2}\boldsymbol{\omega}^T \mathbf{I}\boldsymbol{\omega}$) is met. Even if operations stretch to 1 to 2 weeks, the peak power draw remains well within the capabilities of standard small satellite solar arrays.

4.1.8 TRL & Feasibility

: While space-rated HTS systems and cryo-cooling loops are currently maturing, the fundamental physics of eddy current braking are fully proven and used on Earth in train brakes, emergency stops for industrial equipment, excercise machines, and more. The absence of complex pulse timing, optical targeting, and dependence on unknown surface absorptivity drastically reduces the operational risk compared to laser ablation.

4.1.9 System Robustness

: The system does not rely on target having specific ablative coatings or geometries (as laser ablation cannot detumble smooth spheres). It capitalizes on the ubiquitous use of conductive metals (like aluminum, steel, titanium) in satellite manufacturing. The induced torque is purely a function of the target's relative angular velocity and the applied B field, guaranteeing a damping effect on uncooperative debris regardless of its surface condition.

Due to the unfeasible peak power demands of laser ablation and its inherent reliance on in-situ momentum coupling, the HTS magnetic eddy current de-tumbler provides a far more power-efficient, predictable, and robust solution for the SCRAP architecture.

4.2 GOLD De-orbiter

4.2.1 Overview

The Gossamer Orbit Lowering Device subsystem is responsible for contacting the de-tumbled candidate and attaching a passive drag device that accelerates orbital decay. After the magnetic de-tumbling system reduces the target's angular velocity within mission specifications, the servicing spacecraft deploys the GOLD attachment package using a 6-axis boom. Shown in Figure 8, this package carries an extremely thin gossamer balloon that is later inflated to create a large drag surface, as well as multiple adhesive pads on the bottom surface to secure the system to the candidate object. Once attached, the balloon inflates to increase the effective cross-sectional area of the target so that interaction with the atmosphere in LEO gradually lowers its altitude until re-entry occurs.

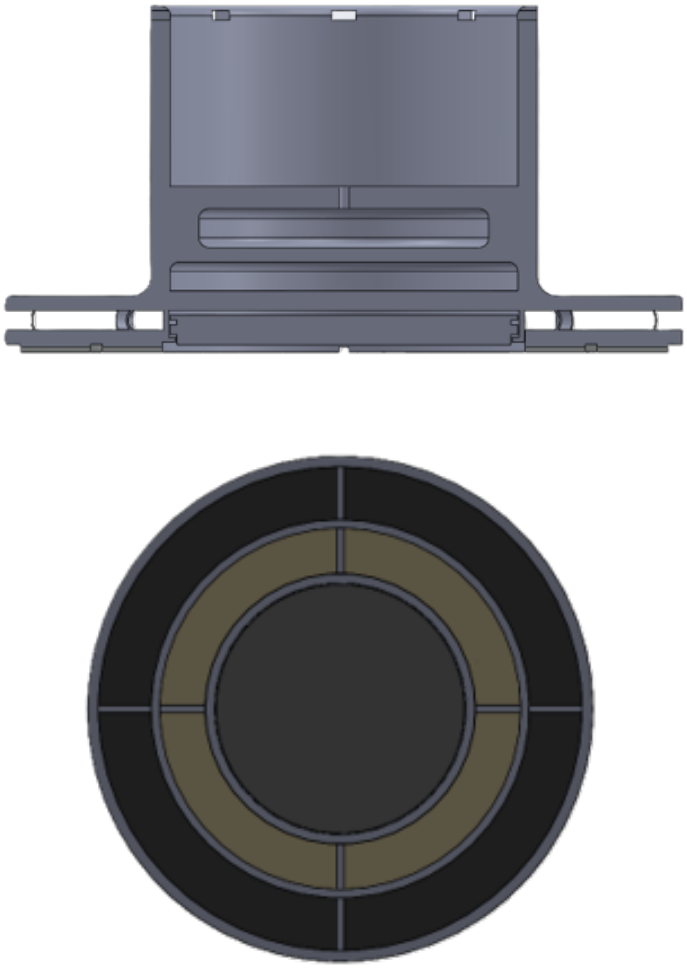


Figure 8: Side and Bottom Views of Gossamer Orbit Lowering Device (GOLD) Attachment Package.

4.2.2 Attachment and Adhesion Method

An attachment package is extended from the satellite carrying GOLD on a boom arm and brought into contact with the target spacecraft surface. Initial contact is achieved using a temporary Van der Waals adhesive, which provides low-strength adhesion while the thermoplastic adhesive activates. The thermoplastic layer is designed to soften and conform to the local spacecraft surface geometry and adhere once heated to approximately 20°C. After the thermoplastic has set, a Jet-Weld adhesive pad is heated to 149°C and pressed against the surface at 14 kPa, producing permanent adhesion with a bond strength of approximately 34 kPa.

A three-stage adhesion mechanism is critical for proximity operations due to the total inertial nature of a space environment. Prolonged candidate-chaser interfacing increases the possibility of damage to both systems, and is necessarily avoided by minimizing contact time. Thus, fast touch-off (on the order of milliseconds) reduces linear momentum transfer and prevents residual angular momentum from significantly affecting chaser orientation and potentially damaging the package or boom. However, the Van der Waals adhesive is incapable of supporting the total drag

force experienced throughout the de-orbit lifetime, and thus secondary and tertiary adhesives are employed to step up to the required bonding strength.

This approach enables attachment in vacuum, since dry adhesives do not face bubbling issues and storage complexity as liquid adhesives do, and allows the system to bond to satellites with irregular surfaces using the deformable thermoplastic and Jetweld, which are unachievable otherwise.

4.2.3 Balloon Inflation and Structural Design

Once the system is attached, an onboard microcontroller triggers an actuator that breaks the seal between the vacuum balloon chamber and a stored solid sublimating material. The material slowly sublimates, producing gas that inflates the balloon to a total volume of 95.5 m^3. At an internal pressure of 50 Pa, the balloon expands to a radius of approximately 2.84 m, significantly increasing the drag area of the debris object. The balloon has a mass of about 3.6 kg and is made from thin Kapton Polyimide film, which was selected for its lightweight strength and thermal stability. A diamond-grid ripstop reinforcement is also incorporated to help the membrane withstand pressure loads and prevent tear growth during operation.

4.2.4 Balloon Sizing and Pressure Requirements

The balloon size is derived from the drag force required to de-orbit the target within the allowable mission timeline. This design process relates the required drag area to the desired de-orbit time, ensuring the system can provide sufficient atmospheric braking once deployed. Shown in Table 1, the selected design yields a reference area of 25.253 m^2 and a balloon radius of about 2.835 m, using a 25 μm Kapton membrane. Material property analysis was used to determine the safe operating range, including upper and lower pressure limits set by rupture resistance and membrane collapse prevention. The final operating pressure of 50 Pa is above the minimum inflation pressure and well below the rupture pressure, allowing the balloon to maintain shape while remaining structurally safe throughout the de-orbit process.

Table 1: Balloon sizing and material properties

Parameter	**Value**	**Unit**
Reference Area	25.253	m^2
Radius	2.835	m
Balloon Thickness	25	μm
Total Material Volume	0.0025	m^3
Kapton Density	1420	kg/m^3
Material Mass	3.58	kg
Ultimate Tensile Strength	231	MPa
Tensile Modulus	2.3	GPa
Poisson's Ratio	0.34	–
Rupture Pressure	814.76	Pa
Minimum Inflation Pressure	0.0382	Pa
Operating Pressure	50	Pa

4.2.5 De-Orbit Procedure

After attachment, the deployed GOLD structure increases the effective drag area of the debris object and passively lowers its orbit through atmospheric drag. Although the atmosphere in low Earth orbit is extremely thin, the added surface area causes a measurable increase in drag force over time, gradually reducing orbital altitude until re-entry occurs. Shown in Figure 9 for example, a 100 kg target satellite starting at 500 km altitude can be lowered to about 120 km in roughly 60 days, after which it will de-orbit naturally. In this way, GOLD provides a lightweight and passive method for post-mission disposal by combining reliable attachment, controlled balloon inflation, and increased aerodynamic drag.

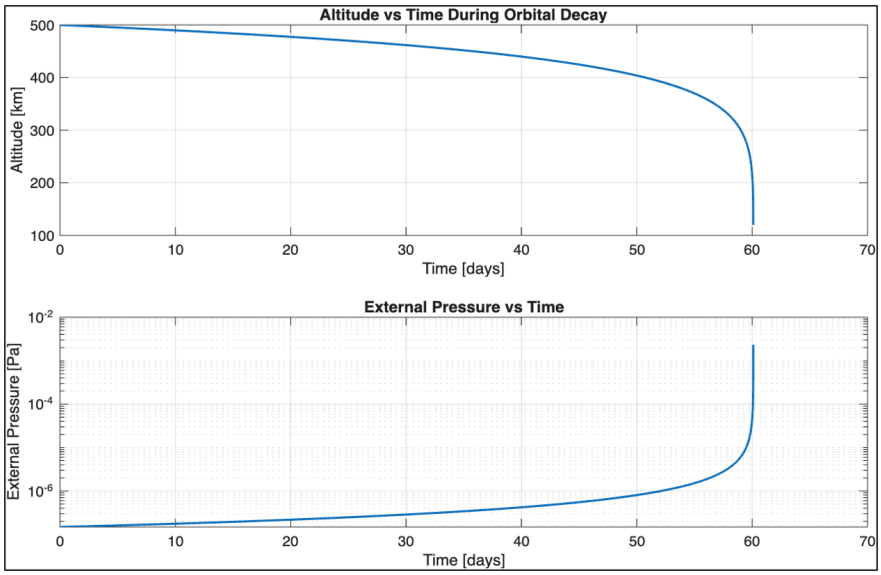


Figure 9: 100 kg Target Example Analysis.

4.3 Bus Components

This subsystem covers all essential parts of the satellite that do not include our Payload (De-Tumble Solution, De-Orbit solution, and any supplemental hardware such as cameras). The following major sub systems are included

- Electrical Power System (EPS)
- Propulsion/ Fluids System
- Flight Computers
- Thermal System
- Communications System
- Navigation System
- Bus Body

Several of these systems are designed around commercial off the shelf components and will be sized around baseline requirements for LEO missions, the only primary exceptions are the Thermal Management, Propulsion, and Power Systems. These systems are being more appropriately vetted to account for the increased thermal, maneuverability and power requirements needed for the payload system to be fully operational.

4.4 Propulsion System

4.4.1 Electric Propulsion System

The Primary mode of propulsion for the satellite will be a single high power Hall Effect Thruster system. It was sized to be capable of preforming station keeping burns, orbit raising between target satellite orbits, and preform rendezvous approach between the chaser and the target satellite. The system selected is a BHT-1500 full hall thruster system from the Busek Co Inc, which contains the following primary components.

- 1 BHT-1500 Hall Effect Thruster
- 1 Composite Over-wrapped Pressure Vessel (COPV)
- 1 BHT-1500 Propulsion Power Unit (PPU)
- 1 Full plumbing system (Feat. mass flow control unit, valves, and tubing)

In addition to the above components which are provided by Busek, scrap will integrate the following component in the system

- 1 Two-Axis Gimbal

This thruster has supposedly not flown on any missions in the past, however it has extensive literature surrounding its developmental and test history. Busek has a strong reputation in the EP community for their thruster reliability and production capabilities, as such it was a natural manufacturer of choice.

The thruster itself is capable of operating in a high thruster or high ISP configuration based on the needs of the satellite at any point in its mission (Ex, orbit raise could be operated in the

high thrust mode while EOL operations could see the high ISP mode be used). Since this satellite will have a constantly changing center of mass because of the gold system, it was decided that the thruster should be placed on a 2 axis gimbal, such that it is always capable of firing directly into the center of mass over the course of the mission.

To minimize the risk of having the charged particles emitted by this thruster interacting with either our spacecraft components or a target satellite, the thruster is placed opposite the face containing the electromagnet and gold systems.

4.4.2 Mono-propellant System

The secondary mode of propulsion for the spacecraft will be a hydrazine mono-propellant system. This system will be responsible for fine pointing of the satellite during rendezvous operations and (if necessary) de-saturating the spacecraft reaction wheels.

The whole mono-propellant system will be provided by the Ariane Group and consists of the following components

- 16 Ariane 1N Hydrazine Thrusters
- 1 DT180 Diaphragm Tank
- 1 Full Plumbing system (Feat. mass flow control unit, valves, and tubing)

All components featured in the mono-propellant system have extensive flight heritage on other legacy spacecraft.

The monopropellant thrusters themselves are placed on the corners of the spacecraft (2 each per corner, for a total of 16). The thrusters themselves can therefore be used to fire through the center of mass when fired in synchrony on a particular face, or to produce a torque on the spacecraft when fired separately. This system gives the spacecraft the ability to maneuver into particular orientations which are ideal for either de-tumbling the target spacecraft or attaching the gold system.

4.5 Communication System

4.6 GNC/ADS System

5 Sizing and Feasibility

5.1 Mass Budget

Mass estimates for the total satellite were determined by summing the total mass of all components that were sized to be included on the satellite. This includes mass from all commercial off the shelf components (PMAD, Propulsion, ect.), In house components (GOLD and De-Tumble Sub Systems), and mass estimates from components that were indeterminately sized (Bus Structure, Thermal Control, ect). A Figure showing the total cost of the spacecraft is shown below.

5.2 Power Budget

Budget values were found by directly researching specific parts and getting an averaged cost for said component, focusing only on critical hardware components with high costs, ignoring costs to assemble or test.

- **Propulsion System**
  - ASCENT monopropellant propulsion system (tank, thrusters, valves, feed system): $1.5M
  - Hall effect thruster propulsion system (thruster, PPU, tank, gimbal): $2.0M
- **Bus Architecture**
  - Power system (solar arrays, PMAD, batteries): $1.8M
  - ADCS & navigation (reaction wheels, star tracker, IMU, GPS): $1.2M
  - Structure & thermal control (radiators, heaters, MLI): $1.0M
  - Avionics & communications: $0.8M
- **Magnetic Detumbling System**
  - Superconducting magnet assembly: $0.8M
  - 6-DOF boom arm / robotic manipulator (G.O.L.D. system): $2.5M
- **G.O.L.D. Capture System**
  - Extendable adhesive capture arms & mechanisms: $1.2M
- **Total Basic Hardware Cost Estimate:** ˜$12.8M *(Note: Original estimate provided was $14.8M; adjusted for sum of parts)*

5.3 ACDS / GNC

Attitude control is very important during de-tumbling since there will be a transfer of angular momentum from the target to the spacecraft. Reaction wheels will be used to stabilize attitude during maneuvers. However, it is still likely that with the large momentum exchange there will be a necessity to dump momentum. A mono-propellant system will work in supplement to the reaction wheels. This is expanded on in more detail in the Propulsion subsection below.

Earth-frame attitude and position determination will be achieved using a star tracker, GPS, and an onboard IMU to propagate between readings. Although this will not be enough to guarantee the accurate position needed to interact with target objects. Two cameras will be used to detect the range, direction, and rotational state of the target.

In between targets, it may be beneficial to dump momentum using magnetic torquers to save fuel. This is still being considered for addition to the spacecraft.

5.4 Propulsion

5.4.1 Electric Propulsion

As mentioned previously, the electric propulsion subsystem was sized under the assumption that it was the primary mode of propulsion for the satellite to preform orbit raises, station keeping, and initial chasing of the target satellite. Our GNC software was therefore used to calculate the expected deltaV it would take to preform orbit raises between target satellites as well as the deltaV it would take to approach the satellite once the chaser is placed in the same orbit. Using this program we were able to determine that an ideal deltaV for this kind of mission would be 1800 m/s. Using this detlaV data requirement, we were able to take the publicly available data on the ISP of several different hall thruster solutions to determine the ideal model and tank size of an

ideal hall thruster system using a MATLAB calculator. We ended up settling on the BHT-1500 from Busek because of its exceptionally high ISP and total impulse without exploding our power budget. The high ISP also gives the satellite significant weight savings on Xenon propellant mass. The Baseline requirements for featuring this propulsion system were found and are displayed below.

- Xenon Mass Needed: 40Kg
- Whole System Dry Mass: 20Kg
- Max Required Power During High Thrust Mode: 2700 W

Mass of the total system was determined by observing historical data for other busek systems and scaling accordingly. Xenon mass was calculated directly for our mission needs based on the nominal ISP of the thruster. The max required input power is derived from the thruster data sheet (**Put Source here**)

5.4.2 Mono-Propellant

As mentioned previously, the Mono-Propellant system is the secondary mode of propulsion for the satellite, and is responsible for small adjustments in satellite position when approaching the target and desaturation of reaction wheels (if necessary).

As displayed in the GNC section, the use of the mono-propellant system for de-saturating reaction wheels will be incredibly limited due to the addition of a magnetometer and large reaction wheels (we find that a burn of less than 1 m/s would be sufficient to counteract the torque generated by de-tumbling objects), however in the event that a particularly large object is de-tumbled the system can prove useful.

Fine pointing of the satellite itself Similarly requires small burns of small delta V, and as such also does not necessitate the incorporation of a large tank. We have been able to size the whole mono-propellant system below accordingly.

- Hydrazine Propellant Mass: 20kg
- System Dry Mass: 40kg
- Max Required Power: 100W

The system dry mass was calculated by observing historic whole system masses using the DT180 tank we specked for the role. The max required power is an estimate driven by the historical power requirements needed to operate half (8) hall effect thrusters at any given time (**Source Needed**)

5.5 Power

- **Power Generation**
  - Solar array generation (BOL): ~ 1.8 kW
  - Solar array generation (EOL): ~ 1.4 kW
- **Energy Storage**
  - Battery capacity: ~ 3 kWh
- **Operational Loads**

  - Estimated spacecraft average load: ~ 800 W
  - Peak operational load: ~ 1.5 kW

- **Subsystem Power Draws**
  - Hall thruster electric propulsion: ~ 600 W
  - Superconducting magnet assembly: ~ 400 W
  - Total payload power requirement: ~ 600 W

This spacecraft will use fairly standard architecture for spacecraft power systems. Namely including the following subsystems

- Solar Panels
- Central Power Managements and Distribution System (PMaD)
- Batteries

Similar to other systems on this satellite, the Power system requirements are determined by the needs of the payload system which are still a work on progress.

The Solar Panel Sub System will be sized based on the power requirements of the system. We intend to baseline a commercial off the shelf solution from oe of several solar array manufacturers (Blue Canyon, Lockheed Martin, Red Wire, ect)

The PMaD and battery systems will similarly be sized based off of the power requirements. Oftentimes these two systems are sold as a joint solution from top of the line vendors (Bradford Space, Endurosat, Ibeos)

5.6 Thermal Control Strategy

The De-Tumble solution on this spacecraft makes use of high temperature semiconductor devices (>70K) which will in all likelihood require an additional cooling system to ensure the semiconductors never go above their quenching temperature. The state of the art solution for keeping components on spacecraft below 70K is a Liquid Helium Cryo-Cooler solution. Commercial off the shelf Cryo-Cooler solutions already exist on the market; However, depending on the needs of our payload system a ground up solution may be needed either in tandem with the cryo cooler or replacing it.

5.7 Communications Link Budget

The communications protocol is not of particular importance due to the desired autonomous nature of the mission architecture. Thus, uplink and downlink of constellation targeting and subsequent de-tumble and de-orbit data is the main concern. The two main candidates for this task are Radio Frequency (RF) and Free Space Optical (FSO) communications. RF is heritage technology and been proven time and time again, meaning it is comparably more reliable and cheaper than FSO systems. FSO, conversely, is more expensive, but allows for much higher data rates. However, considering that RF is also more reliable in communications during bad weather (FSO is highly affected by diffraction effects caused by weather patterns such as clouds), it makes the most sense to continue with RF communications systems.

6 Cost and Scheduling

6.1 Cost of Components

The cost of components for the spacecraft is defined as the cost of producing a single satellite, including the cost of AI&T but not including the cost of R&D since those values would likely be very inaccurate considering the current stage of development

This satellite was design featuring a combination of Commercial off the shelf solutions and in house manufactured components. Most commercial off the shelf components could be priced using a variety of online sources. In house component costs were calculated by taking the cost of the raw materials required to produce a unit. To account for the cost of processing and other small components not specked for the mission, an overhead margin of +30% was added to the total cost of the satellite. The total cost chart is shown below

6.2 Schedule for Profitability

7 Risk Assessment & Feasibility

7.1 Major Technical Risks

As with any complex system, particularly novel technology configurations, technical risks comprise the majority of mission failure modes. Identifying and categorizing risks are critical in understanding possible deviations in projected performance and fully characterizing system capabilities. These risks are visualized within a traditional risk matrix, which allows proper care and focus to be placed on high criticality risks. They are further clarified in descending order of criticality.

7.1.1 Magnet De-Tumbler

Primary risks associated with the magnetic de tumble concept have to do with the electromagnetic coils themselves. The largest immediate risk is super conductor quenching, the process by which a superconducting material suddenly and irreversibly loses its superconducting properties. In the case of YBCO (our semi conductor solution) this will cause the electromagnet to cease function and render the entire system useless. This requires us to maintain the following conditions from the manufacturer specifications.

- Keep coil temperature below $\sim 77^{\circ}$ K during operation
- Keep current density (J_c) below manufacturer specifications
- Keep magnetic field density (H_c) below manufacturer specifications

The Secondary risk of the electromagnet is the fact that magnetic fields generated by the magnet will be bi-directional and run the risk of interacting with components on the satellite bus.

7.1.2 GOLD System

The primary risk associated with the GOLD subsystem is risks associated with collision. This is the case because the GOLD system must be physically attached to the satellite using a boom. Unwanted collisions with the candidate satellite run the risk of generating more debris from chipped pieces or undoing the de-orbiting process from the magnet coils

The secondary risk for GOLD is the solution detaching during de orbit. The de-orbit process increases the drag force on the system by increasing the effective surface area of the spacecraft. The force generated by this must be less than the maximum total force that the Van Der Waal's bonds can sustain.

7.2 Mitigation Strategies

After ranking risks by criticality, it is also important to determine sufficient mitigation strategies to reduce the likelihood and/or impact of an event occurring, either by implementing redundant systems, operational constraints, redline sensors, etc.

7.2.1 Magnet De-Tumbler

The Primary method to reduce risk of quenching for the Superconductor is to ensure that the coils are kept below 92K via the use of an effective solar insulator to protect the coils from baking in the solar radiation.

To mitigate the magnetic field strength emanating from the electromagnet back onto the spacecraft, it is possible to surround the back side of the electromagnet with MuMetal, a specially designed alloy that is able to curve magnetic field lines away from sensitive components. This product has extensive flight heritage

7.2.2 GOLD System

To mitigate the risk of unnecessary collision between the GOLD boom and the candidate satellite there will need to be extensive GNC and Robot vision based testing before flight with the boom arm. This will ensure that collisions are minimized

The risk of the GOLD system becoming detached during orbit has several potential mitigation strategies. Firstly, the attachment surface on the GOLD system will need to be remotely inspected and approved for maximal connection integrity. The ideal attachment surface minimizes surface roughness, which could be determined using a simple LIDAR system.

7.3 Feasibility Summary

Many aspects of both the GOLD system and the Magnet De-Tumble System are still getting properly vetted. For example, the current density in the electromagnet we have designed is below the quenching current density, so if we can create a stable thermal environment for the coil to inhabit we may be able to do without a dedicated cryo-cooler system. This is however still being vetted and may change. In order to more thoroughly determine this, we will continue to develop sims for our electromagnet with more advanced parameters. The GOLD system is a bit behind the Magnet de-tumbler, as a result the risks and mitigation strategies were based off of less matured results. In order to change this we will similarly continue to develop the GOLD system (create more effective cad, select a specific adhesive solution, determine the exact effects of surface roughness on force strength, ect)

8 Communications and Data Handling

The spacecraft uses an S-band link to support reliable command, telemetry, and payload data exchange throughout the mission. A turnstile antenna is used for mission telemetry and health packets, while a parabolic dish enables higher-volume downlink of mission data. Normal operations

are handled through a private ground station, with AWS Ground Station support available during critical mission phases for extra reliability.

Communication is organized using CCSDS packet standards, allowing uplink commands such as proceed, wait, maneuver, and request. These commands let operators authorize mission steps, place the spacecraft in a power-safe mode for analysis, command orbital maneuvers, or request specific data products. On the downlink side, the spacecraft returns health data such as bus voltage, solar array current, temperature, and fault flags, along with telemetry containing health information and time-history mission data, full mission logs, and video of the target when needed.

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